

This homework assignment is worth 25 points total.

Answer the following questions to the best of your ability, writing your answers by hand on white paper. Don't write too small, and use clear handwriting. You may use a pencil or pen, but not one that uses red ink.

You will need to submit your solutions both physically and digitally. The digital copy must be submitted to Moodle by Tuesday, February 25 by 23:59. The hard copy is due on Wednesday, February 26 just before the start of your lecture section. The physical copy must be identical to what was submitted digitally.

Scanning Guidelines:

- Make a good quality scan. Phone pictures (CamScanner) are acceptable, as long as they are in focus and readable.
- Before submitting, verify that you can read your own scans.
- Scan the pages in the correct orientation (not rotated over 90 degrees.)
- You may submit your scans in a single .pdf file with the solutions presented in the same order as the questions, or collect them into a single archive file (.zip, .gzip, or .rar) with your files numbered in such a way that they appear in the right order.

1. (12 points total; 2 points each) Determine whether the following languages are:

- Regular
- Context free, but non-regular.
- Non-context free

If regular, provide a DFA, NFA, or a regular expression for the language.

If context free but not regular, provide a grammar or PDA for the language AND justify your claim that it is non-regular by providing a word w dependent on p that could be used in a proof utilizing the pumping lemma for regular languages.

If non-context free, justify your claim by providing a word w dependent on p that could be used in a proof utilizing the pumping lemma for context free languages.

- $\{a^n b^{2m} \mid n \geq m\}$
- $\{a^k \mid k \text{ is a Fibonacci number}\}$
- $\{v \in \{a, b, c\}^* \mid \text{the number of } a\text{'s in } v \text{ equals the total number of } b\text{'s and } c\text{'s in } v\}$
- $\{a^n b^{2n} a^n \mid n \in \mathbb{N}\}$
- $\{s\#t \mid s, t \in \{a, b\}^*, \text{ where } s \text{ is a substring of } t\}$
- $\{sts \mid s, t \in \{a, b\}^*\}$

2. **(2 points total)** Consider the language with terminal alphabet $\{a, b, c\}$ generated by the following grammar:

$$S \rightarrow S a b S$$

$$S \rightarrow c$$

- a) Is the above grammar ambiguous? Justify your answer.
- b) Is the given language regular? Justify your answer.

3. **(3 points total)** Consider the language with terminal alphabet $\{a, b\}$ generated by the following grammar:

$$S \rightarrow b S S$$

$$S \rightarrow a$$

- a) What is the relationship between the number of a's and the number of b's for any given word in the language? Justify your answer.
 - b) Is the above grammar ambiguous? Justify your answer.
 - c) Is the given language regular? Justify your answer.
4. **(2 points total)** Using the construction from the slides, create a Generalized PDA (GPDA) that accepts the language recognized by the below grammar using the top-down translation approach. Be sure to use the same notation as we used in class and in the slides.

$$S \rightarrow S A a$$

$$S \rightarrow a b$$

$$A \rightarrow b A$$

$$A \rightarrow a$$

5. **(4 points total; 2 points each)** Suppose we have the following language:

$$L = \{ a^j b^k c^m \mid k = j + m \}$$

- a) Create a context-free grammar with five rules which recognizes L.
- b) Create a pushdown automaton which recognizes L.

6. (2 points total) Create a context-free grammar with seven rules which recognizes the same language as the following pushdown automaton.

